



Gurinder Singh

Staff Software Engineer (Mobile & AI)

LLM Agents · RAG Pipelines · Mobile · Automation

Portfolio

LinkedIn

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SCAN FOR PORTFOLIO

90M+

USERS IMPACTED

28+

HACKATHONS

2×

1ST PLACE WINS



PATENT HOLDER

SUMMARY

Staff Software iOS/Mobile & Applied AI Engineer with over a decade of experience architecting and scaling production consumer platforms and AI-powered systems used by millions. Patent holder and technical leader with deep expertise in iOS platform architecture, distributed client systems, and AI-enabled product development—focused on translating emerging AI capabilities into reliable, real-world products.

Proven track record of taking ideas from early prototypes to globally deployed products, spanning startup innovation through enterprise-grade systems serving 90M+ users, most recently at PayPal/Venmo. Experienced in building agentic AI systems, RAG pipelines, and LLM-powered workflows alongside large-scale mobile features and strategic partnerships, with a strong product mindset and comfort operating in ambiguous, cross-functional environments.

Brings a human-centered perspective to applied AI, emphasizing the elevation of human skill, judgment, and ethos through careful system design. Actively invested in mentoring engineers, shaping technical culture, volunteering with NYC high schools and nonprofits, and building products grounded in context, nuance, and real-world use at scale.

TECHNICAL SKILLS PORTFOLIO * Condensed overview — full technical depth available on portfolio

CATEGORY	SPECIALIZED DOMAINS	TOOLS & TECHNOLOGIES	KEY TECH
Mobile	iOS Core Spatial & Hardware Cross-Platform	Swift, Objective-C, SwiftUI, UIKit, SwiftData, Core Data, ARKit, RoomPlan, BLE, NFC, MapKit, CoreML, MLX, RxSwift, Combine, Flutter & Dart	
AI/ML	Models & LLMs Orchestration Media & Vision	Python, PyTorch, OpenAI, Anthropic, Gemini, Llama LangChain, LlamaIndex, CrewAI, N8N, MCP Servers Stable Diffusion, YOLO, OpenCV, Whisper, ElevenLabs	
Cloud & DevOps	Infrastructure BaaS & CI/CD	AWS (Lambda/EC2), GCP (Vertex AI), Azure, Docker Supabase, Firebase, GitHub Actions, Jenkins, Fastlane	
Data & Tools	Databases API & Search Workflow	PostgreSQL, MongoDB, Pinecone, ChromaDB, Redis, GraphQL, Figma, Jira, TDD, Agile, Cursor, Xcode	

PROFESSIONAL EXPERIENCE



PayPal / Venmo

2020 – Present

Staff iOS Engineer → Applied AI/ML Engineer



AI/ML Evolution & Technical Innovation (2023–Present)



Project Mercury — AI-Powered Agentic E-Commerce Platform

June 2025 → Present

- ▶ Built a fully functional prototype in 72 hours; gained momentum across org
- ▶ Presented to the SVP of AI, earning immediate backing and leading a high-velocity team
- ▶ Trained five SDXL LoRA models; achieved **23% loss reduction** and 109MB optimized models

AI/ML Implementation & Automation (2024)

- ▶ Developed AI-powered crash detection system using RAG pipelines + Vector stores with automated DataDog monitors — **now adopted team-wide**
- ▶ Created **10%+ productivity improvements** via AI-powered automation tools
- ▶ Championed AI coding assistants across teams, training colleagues on leveraging AI



iOS Engineering Excellence & Business Impact

Major Product Launches & Partnerships

- ▶ **Pay With Venmo Checkout (2024–Present):** Reduced end-to-end pay sheet render latency by ~2 seconds
- ▶ **Venmo Gift Cards (2023):** Led iOS architecture and delivery for Venmo Gift Cards, supporting major merchants like Amazon and Starbucks
- ▶ **Venmo × Amazon (2022):** Built sandbox/integration app in two days for partnership launch
- ▶ **QRC Touch-Free Payments (2021):** Led moonshot initiative winning **\$60M CVS contract**
- ▶ **Amex Partnership (2020):** Built payment-splitting integration for **\$400M partnership deal**

iOS Innovation & Patents

- ▶ **QRC Widget Patent (2022):** Filed "Interface Widget Tool for Automatic QR Code Generation" (PayPal Ref. OCP.D2021.103924.US1). First team to use SwiftUI for iOS widgets at Venmo. **Reduced user friction by 66%**
- ▶ **Dynamic Island / Live Activities / App Clip:** Built end-to-end infrastructure including HLDs, LLDs, and APNS notification systems
- ▶ **Visual Snapshot Testing:** Built comprehensive visual snapshot testing suite using XCTest for UI regression testing



Technical Leadership & Impact

- ▶ Partnered with Design Systems to create modern SwiftUI component library that **cut feature development time by 30%**



Google Stadia

2019 – 2020

iOS Developer/Consultant

- ▶ Worked on Google Stadia for mobile platforms using Flutter/Dart and native iOS plugins
- ▶ Implemented Core Bluetooth integration for Stadia controller connectivity and low-latency communication
- ▶ Utilized Xcode Instruments for performance profiling and CPU/GPU optimization



Morgan Stanley

2018 – 2019

iOS Developer/Consultant

- ▶ Worked on Hybrid (Cordova + Native) Banking and Wealth Management apps for iOS
- ▶ Developed key features: check-deposit, biometric authentication, certificate-pinning, jailbreak detection, push authentication
- ▶ Migrated codebase from 98% Objective-C to 80% Objective-C / 20% Swift

HACKATHON ACHIEVEMENTS & INNOVATION ** Featured highlights — full hackathon history on portfolio*

28+ hackathons attended across 6+ countries as participant, mentor, and judge. Earned 2 first-place wins and served as judge at Yale (2×). Recognized globally for innovation and technical excellence.

1ST PLACE

MLH @ Bloomberg London (2017)
Automated Search & Rescue System.
Featured in BBC documentary.

1ST PLACE

Devcamp NYC (2016)
Talent Management App. Won among 300+ participants at Google NYC.

FINALIST

HackZurich (2018)
AR Car Buying Tool. Finalist among 400+ participants.

Hack Yale – Judge (2×)
Yale University

PenApps – Mentor
University of Pennsylvania

SELECT PERSONAL PROJECTS ** Curated highlights — explore 20+ projects on portfolio*

FEATURED

Ethereal Dimension
AI/ML & Spatial Computing
Austin DoT Partnership:
Built CV system for real-time barricade detection & compliance.
Hylios:
AR room scanner using RoomPlan/ARKit for 3D spatial intelligence. Available on App Store.

Computer VisionYOLOARKitRoomPlanCoreML

FEATURED

VectorVerse Evolve
AI Automation & SEO
Content Pipeline:
Fully automated social media platform with n8n workflows & Strapi CMS. 3-stage pipeline: Intelligence → Ideation → Generation.
Agentic SEO:
AI-powered SEO strategist with keyword research, geographic targeting & magic reports.
Open-WebUI:
Built custom deployment for Azure AI Foundry integration.

n8nStrapiNext.jsAzure AIOpen-WebUI

FEATURED

Rosicrucian Knowledge Explorer
AI Search & Media Platform
Full-stack RAG search engine for 73K+ vectors from digests, books, symposiums, podcasts, YouTube videos.
PDF viewer with page-jump & highlighting, video and podcast player with timestamp seeking. Qdrant + OpenAI + R2. 60-test suite. Amplitude analytics.

Next.jsQdrantR2OpenAISupabase

EDUCATION

BS in Computer Systems (Deferred)
New York City College of Technology, Brooklyn

2012 – 2016

LEADERSHIP & COMMUNITY

Advisory Board Member for multiple tech startups on iOS architecture and AI/ML implementation

Interfaith ERG Chapter Lead (Sikh community): Organized events, secured speakers, created [Gurdwara Langar Seva](#) game to teach Sikh Seva Traditions